

Employee Performance & Productivity

Full Workforce Analytics — Findings, Models & Recommendations

100,000 employees · 9 departments · 7 job titles · hires 2014 – 2024
Automated analytics pipeline · September 2026

Agenda

- 1 Executive summary & KPIs
- 2 Workforce composition
- 3 Compensation & pay equity
- 4 Performance analysis
- 5 Attrition & predictive models
- 6 Workload, remote work & balance
- 7 Key takeaways & recommendations
- 8 Methodology

Executive Summary

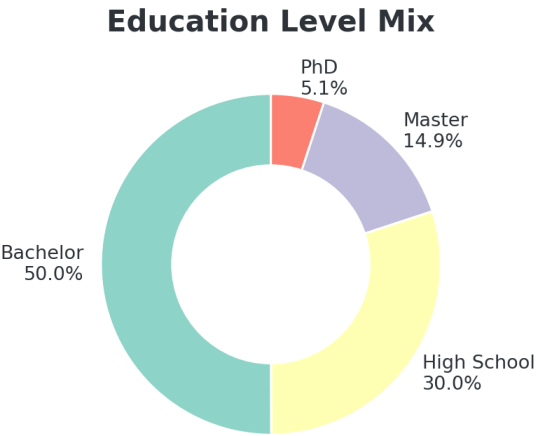
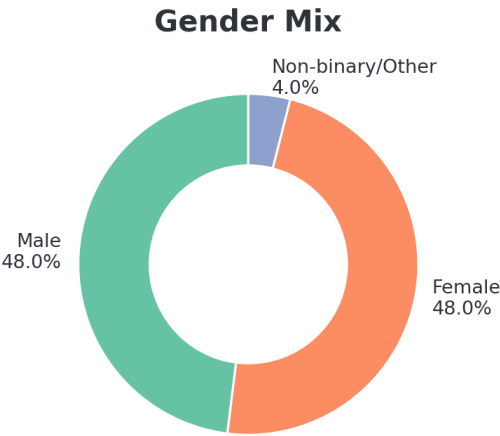
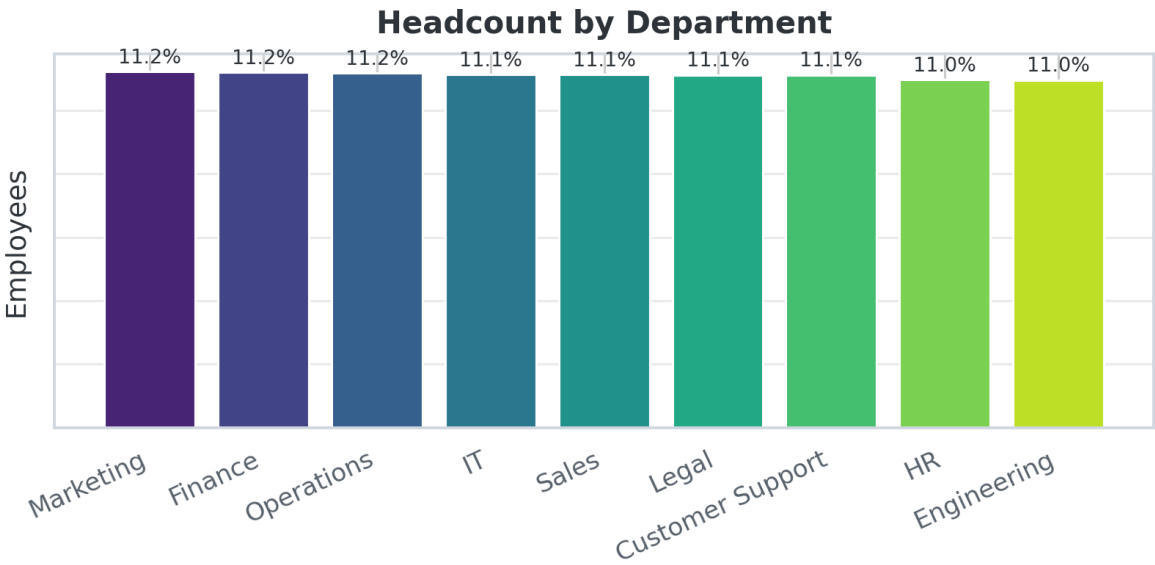
Six numbers and one surprising negative result



- ✓ Pay equity is healthy — women earn \$4.50/mo more on average; within-title gaps $\leq \pm\$25$ ($<0.4\%$).
- ✓ Salary is a formula: job-title premium ($\pm\$2,600$) + \$493 per performance point — model $R^2 = 0.99$.
- ✗ Attrition (10.0%) is statistically unpredictable — both ML models score $AUC \approx 0.49$ (coin flip); every segment sits within 9.5–10.3%.
- ✗ Engagement levers (training, overtime, remote work, team size) show zero measured effect: $|r| < 0.01$ vs performance & satisfaction.
- Departments are near-identical on every metric (salary spread \$39, attrition spread 0.9 pt) — consistent with synthetic data.

Workforce Composition

Evenly balanced across departments, gender and education

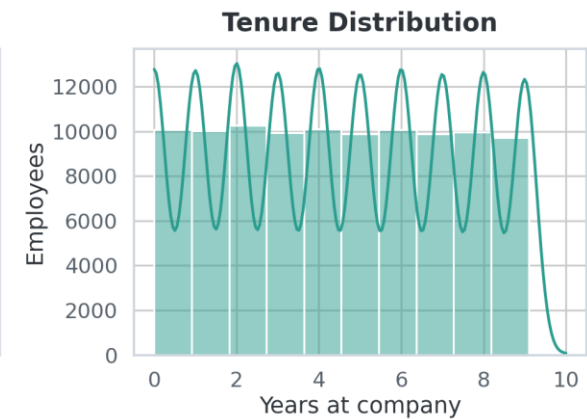
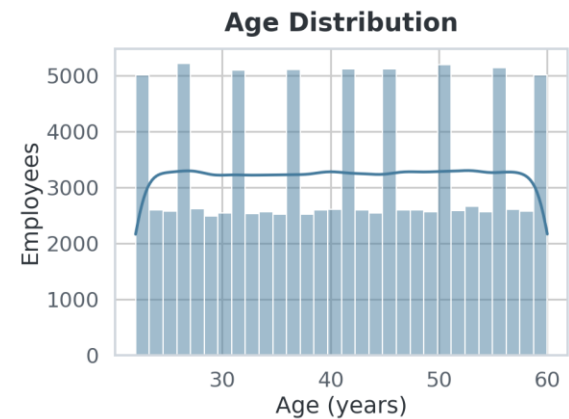
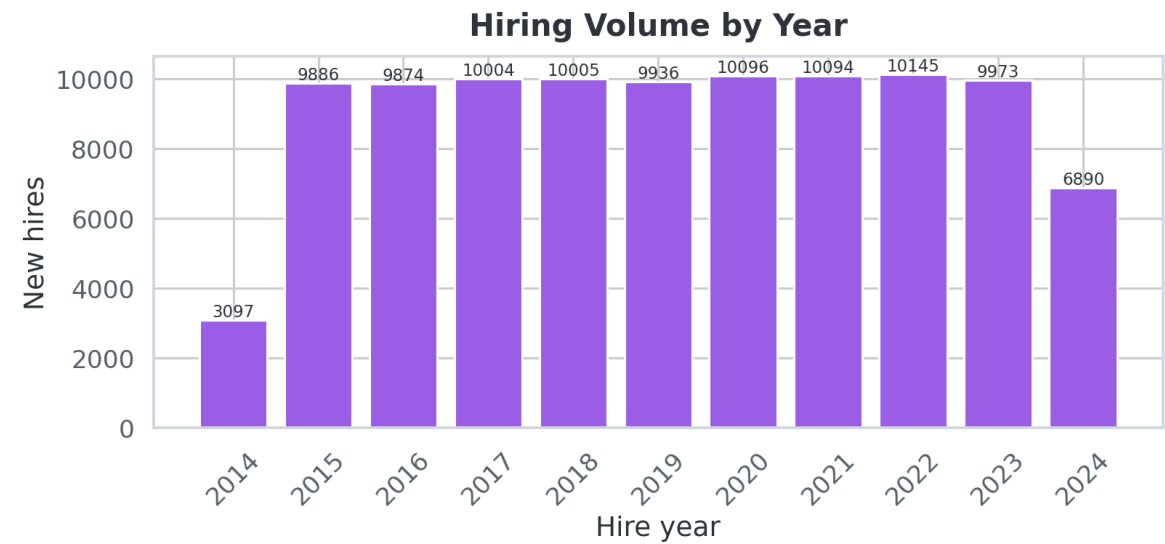


All 9 departments employ 10,956–11,216 staff (11% each).

Education skews to Bachelor; ~9,300 hires/year over the decade.

Hiring Trends & Age Profile

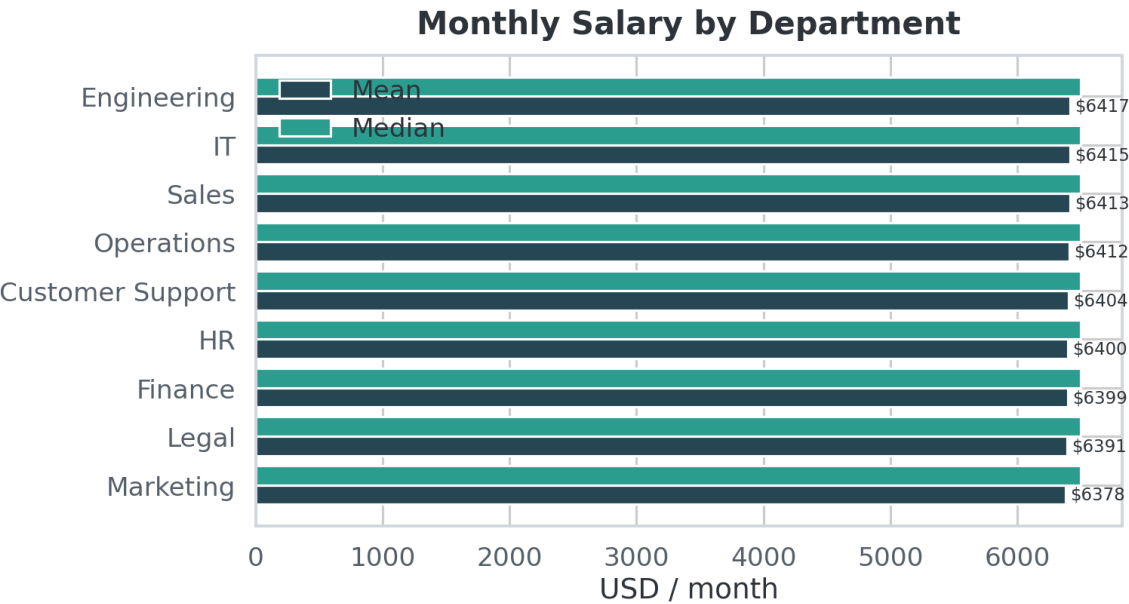
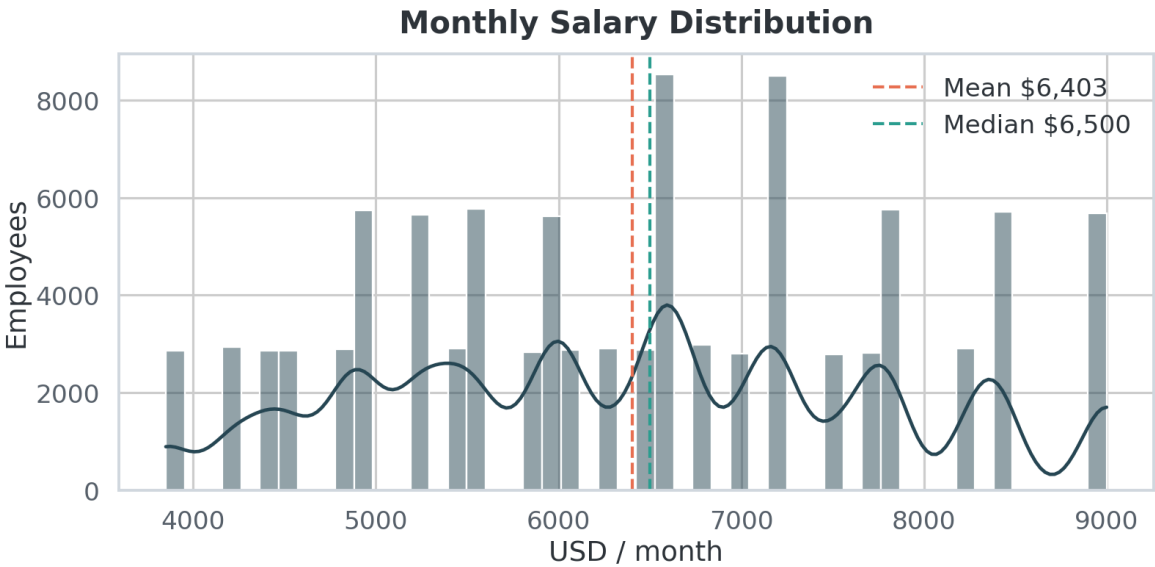
Steady hiring 2014–2024; age peaks in the 40s



Average age 41 · average tenure 4.5 years — a mature, stable workforce profile.

Compensation Overview

Mean \$6,403 · median \$6,500 · σ \$1,373

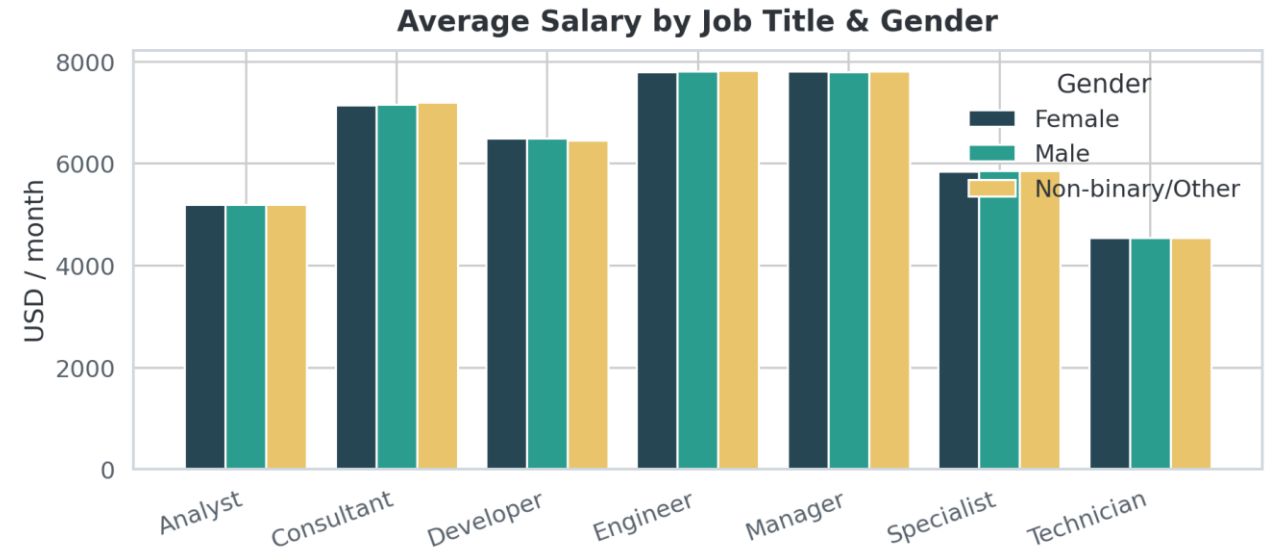
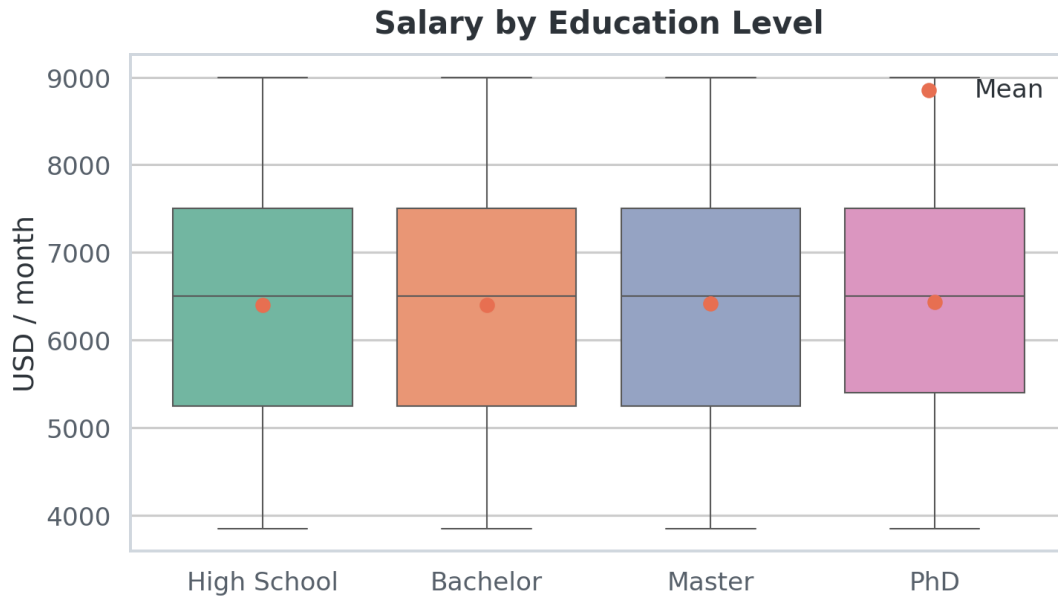


Near-normal salary distribution, \$3,850–\$9,000.

Departments differ by just \$39/month on average — pay is not department-driven.

Education & Gender Pay Equity

Small education premiums · no gender gap to remediate



Company-wide: women +\$4.50/mo vs men. Largest within-title gap: Consultant +\$23 (0.3%).

Verdict: pay equity is healthy — institutionalize quarterly monitoring.

What Drives Salary — OLS Model

$R^2 = 0.991$ · MAE = \$95 on 20% hold-out — pay is essentially a formula

Job title: Engineer **+\$2,600 /mo**

Job title: Manager **+\$2,599 /mo**

Job title: Consultant **+\$1,951 /mo**

Job title: Developer **+\$1,300 /mo**

Job title: Specialist **+\$650 /mo**

Job title: Technician **-\$648 /mo**

Performance score, per point **+\$493 /mo**

Dept, gender, education, age, tenure... **≈ \$0 ($|\beta| < \3)**

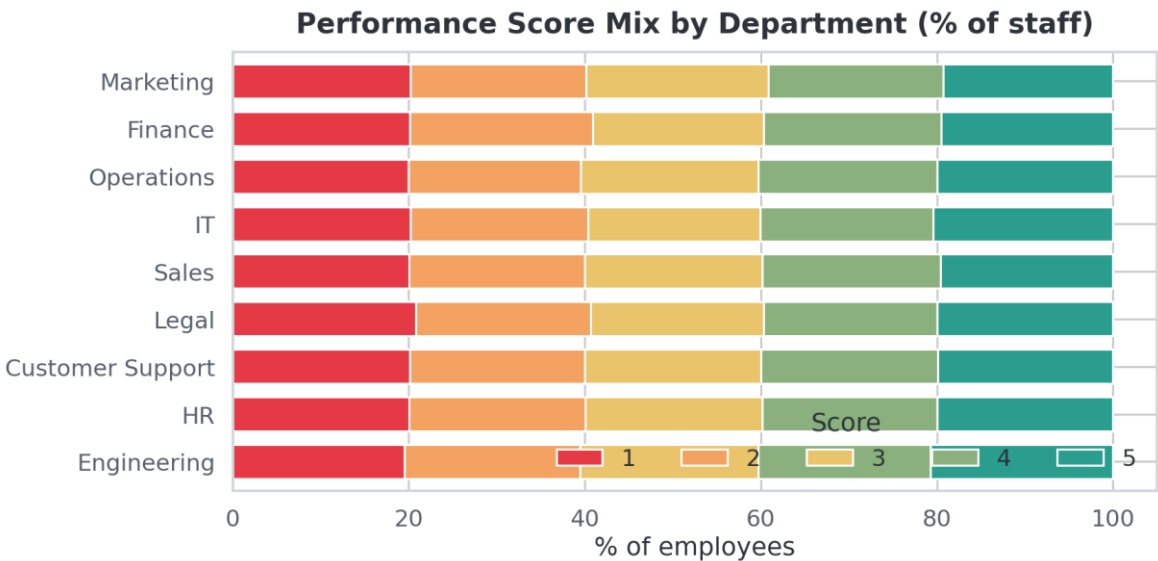
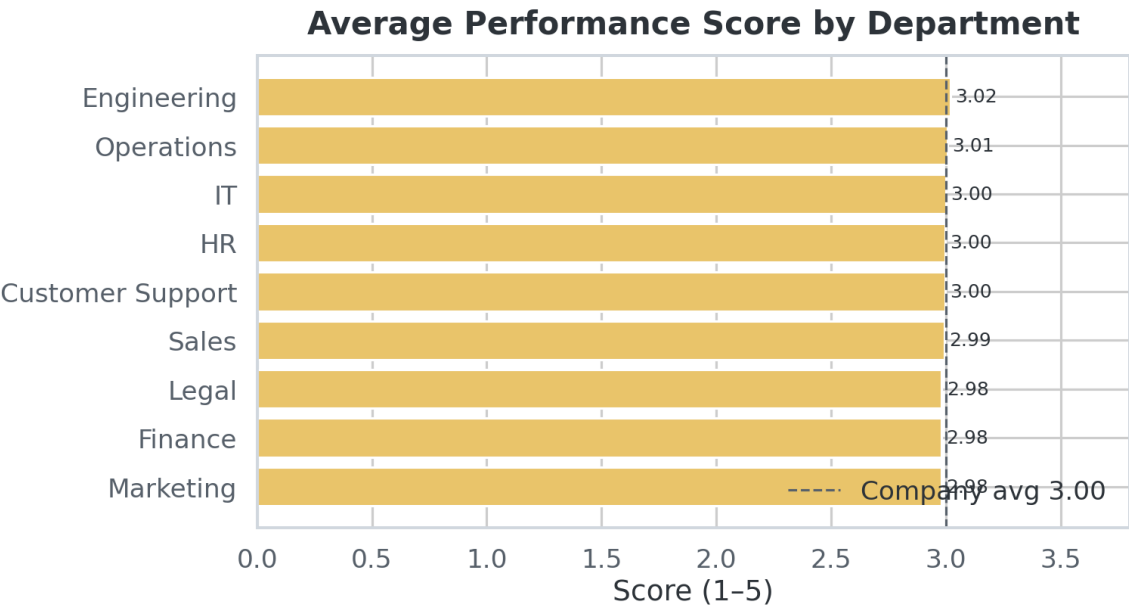
Pay governance is transparent and rule-based — good for fairness.

But performance ratings now directly move pay by up to \$1,972/mo (score 1→5) — rating quality is a compensation-control issue.

Score 1–5 maps to ≈ \$4,430–\$6,400/mo for a Specialist; titles shift the whole curve.

Performance Analysis

Uniform 1–5 distribution · department averages 2.98 – 3.02



39.9% high performers (4–5) vs 40.1% low (1–2) — a perfectly symmetric spread, unusual for real ratings.

Satisfaction & Training vs Performance

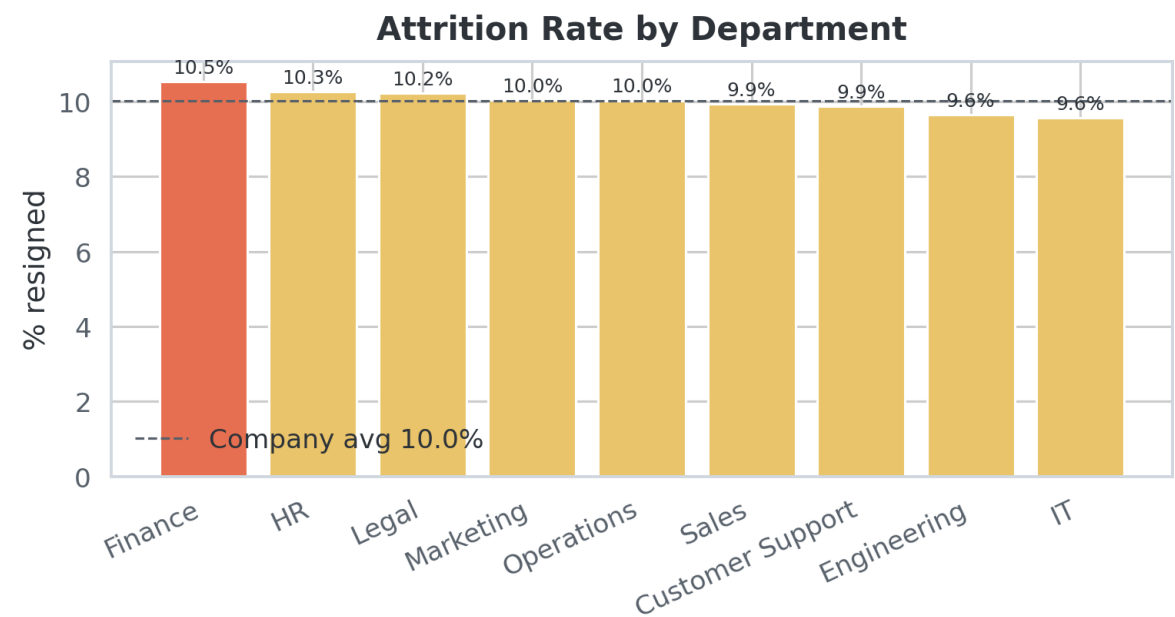
No gradient anywhere — a red flag about measurement



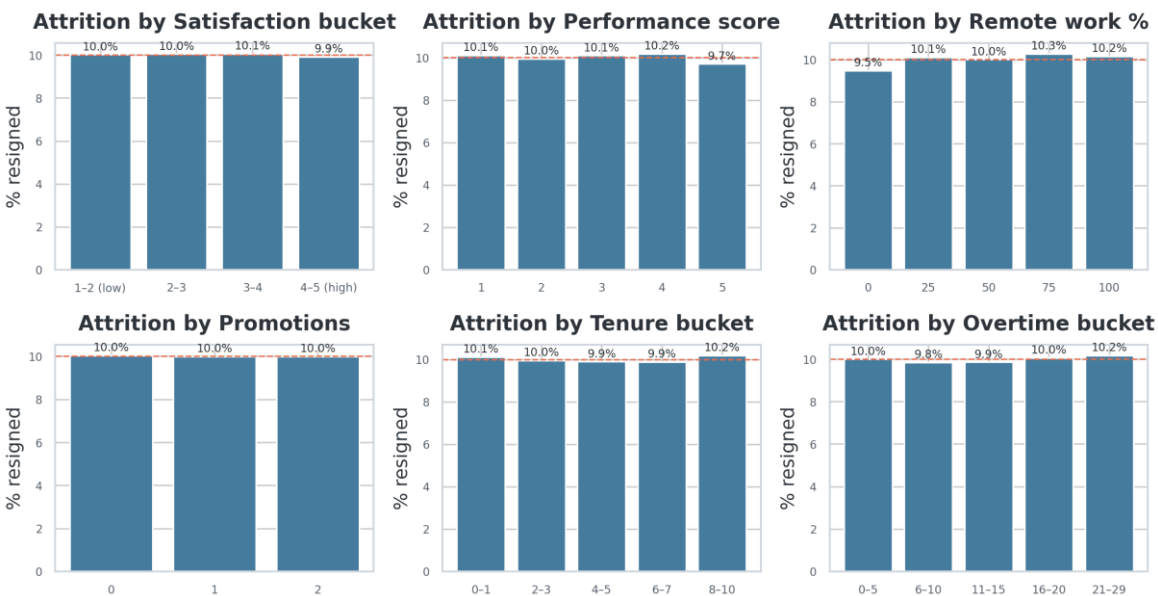
Satisfaction $\perp$ performance ($r = 0.002$) and $\perp$ attrition; training hours (0–99) don't move performance at all.

Attrition Overview

10.01% resigned (10,010) — remarkably flat across the organization



Attrition Drivers — Rate vs. Company Average (dashed line)

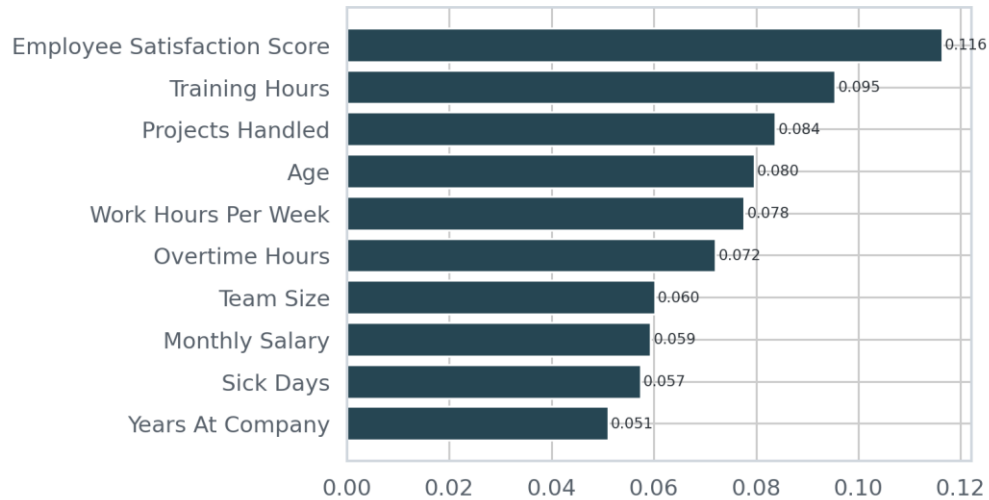


Widest department spread: Finance 10.5% vs Engineering/IT 9.6% — well within noise.

Predicting Attrition — An Honest Negative Result

Both models discriminate no better than a coin flip

Top Attrition Predictors — Random Forest Importance



0.49

Logistic AUC

0.49

Random Forest AUC

0.90

Accuracy = majority-class baseline

All coefficients within ± 0.033 ; all feature importances < 0.002 .

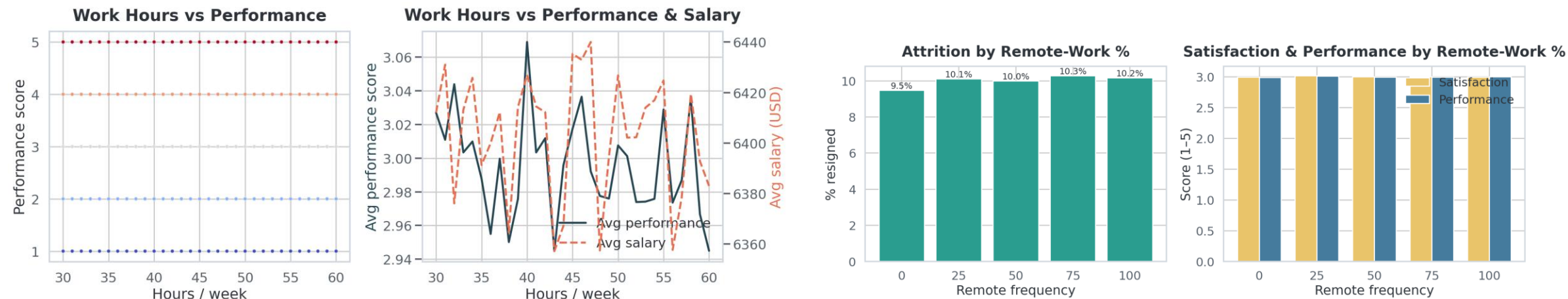
Every segment (satisfaction, performance, tenure, remote, overtime) sits within 9.5–10.3% of the 10.0% rate.

⇒ No flight-risk model is buildable from current fields.

⇒ Action: audit the Resigned flag, capture exit-interview themes & manager ratings, then re-run this pipeline.

Workload, Remote Work & Balance

45h weeks · 14.5h overtime — and no measurable cost or benefit



Attrition is 9.5–10.3% at every remote-work level; overtime and sick days show no burnout signal.

Key Takeaways & Recommendations

1

Protect pay equity

Gaps < 0.4% everywhere. Automate the quarterly audit (already scripted).

2

Audit performance measurement

Scores drive \$493/mo per point yet correlate with nothing — verify they are evidence-based.

3

Fix attrition data capture

AUC 0.49 means leavers are invisible. Validate the flag; add exit themes & manager scores.

4

Pilot, don't blanket-invest

Training/overtime/remote levers show zero effect — test with controlled pilots.

5

Automate the monitor

The pipeline rebuilds everything (charts → dashboard → report → deck) in < 1 min. Schedule it.

Methodology & Reproducibility

Data: 100,000 × 20 CSV — zero nulls, zero duplicate IDs, no cleaning required.

EDA: pandas 2.2 · 20 publication charts in matplotlib/seaborn · Pearson correlations.

Models: scikit-learn 1.6 — LogisticRegression (C=0.5, scaled) & RandomForest (250 trees, depth 8) for attrition; OLS for salary. 80/20 stratified split, seed 42.

Deliverables generated by one pipeline: charts/, dashboard.html (offline, Chart.js inlined), report.pdf (16 pp), presentation.pptx, insights workbook (xlsx), README for GitHub.

Rebuild: python3 src/analysis.py → build_dashboard.py → build_report.py → build_pptx.py → build_workbook.py

Caveat: feature independence + near-perfect salary formula indicate synthetic data — treat results as a validated methodology demo.